Node Link Network Optimization Report

Model Type: **node_link**

Metric	Value
Status	Optimal
Objective Value	$830.00
Total Revenue	$830.00
Total Expansion Cost	$0.00
Total Installation Cost	$0.00
Total Cost	$0.00
Net Profit	$830.00
Accepted Demands	6 / 6
Rejected Demands	0 / 6
Acceptance Rate	100.0%
Nodes	8
Edges	12
Total Initial Capacity	280.00
Total Expansion	0.00
Capacity Increase (%)	0.0%

Insights & Highlights

Key Insight	Detail
Most Congested Edge	R3 -> D2 (88.0%)
Number of Highly Utilized Edges (≥90%)	0
Most Capacity Expansion	None
Total Lost Revenue	$0.00
Number of Expanded Edges	0

Accepted Demands & Flow Paths

Demand 0: S1 -> D1

Bandwidth: 10.0 units **Revenue:** $150.0
Revenue/BW: $15.00/unit

Path	Fraction	Bandwidth
S1 -> R2 -> R4 -> D1	100.00%	10.00

Demand 1: S1 -> D2

Bandwidth: 8.0 units **Revenue:** $120.0
Revenue/BW: $15.00/unit

Path	Fraction	Bandwidth
S1 -> R1 -> R4 -> D2	100.00%	8.00

Demand 2: S2 -> D1

Bandwidth: 12.0 units **Revenue:** $180.0
Revenue/BW: $15.00/unit

Path	Fraction	Bandwidth
S2 -> R2 -> R3 -> D1	100.00%	12.00

Demand 3: S2 -> D2

Bandwidth: 15.0 units **Revenue:** $200.0
Revenue/BW: $13.33/unit

Path	Fraction	Bandwidth
S2 -> R3 -> D2	100.00%	15.00

Demand 4: S1 -> D1

Bandwidth: 5.0 units **Revenue:** $80.0
Revenue/BW: $16.00/unit

Path	Fraction	Bandwidth
S1 -> R1 -> R3 -> D1	100.00%	5.00

Demand 5: S2 -> D2

Bandwidth: 7.0 units **Revenue:** $100.0
Revenue/BW: $14.29/unit

Path	Fraction	Bandwidth
S2 -> R2 -> R3 -> D2	100.00%	7.00

Rejected Demands

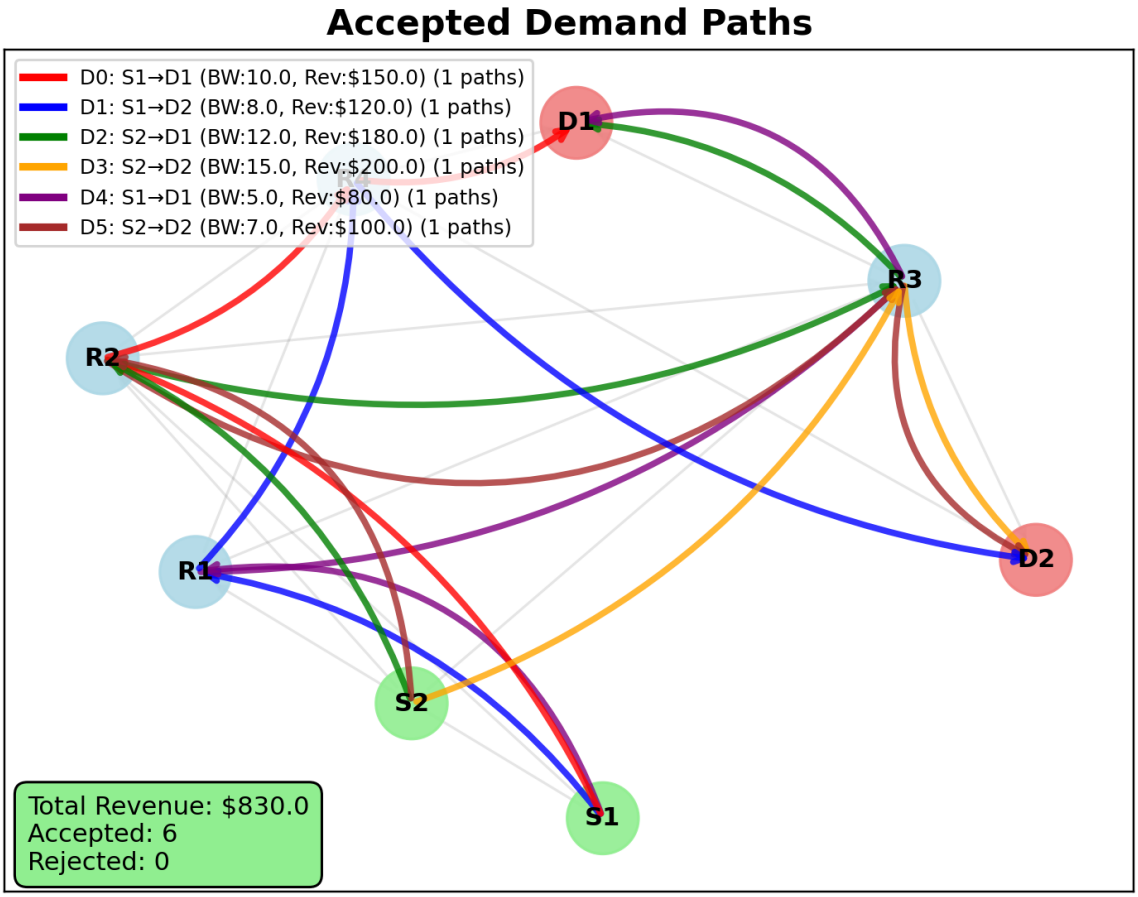
No demands were rejected.

Edge Utilizations (Top 12 Edges)

Edge	Flow	Total Capacity	Utilization (%)
R3 -> D2	22.00	25.00	88.0 %
R2 -> R3	19.00	25.00	76.0 %
S2 -> R3	15.00	20.00	75.0 %
S1 -> R1	13.00	20.00	65.0 %
S2 -> R2	19.00	30.00	63.3 %
R3 -> D1	17.00	30.00	56.7 %
R2 -> R4	10.00	20.00	50.0 %
R4 -> D1	10.00	20.00	50.0 %
S1 -> R2	10.00	25.00	40.0 %
R1 -> R4	8.00	20.00	40.0 %
R1 -> R3	5.00	15.00	33.3 %
R4 -> D2	8.00	30.00	26.7 %

Network Analysis Visualizations

Accepted Demand Paths



Edge Utilization

